



# Gati-Tech AMS

Automobile Dealership Asset Management Platform

## Comprehensive QA & Security Test Report

Full-stack coverage: VAPT remediation · Unit tests · Regression suite

**Report Version:** v2.0

**Date:** June 2026

**Platform:** Next.js 15 / App Router

**Environment:** Staging + Static Export

**Prepared by:** QA Engineering Team

### Executive Summary

All test suites executed against commit HEAD — June 2026 build

76

TOTAL  
TESTS

76

PASSED

0

FAILED

8

VAPT  
FIXED

97%

COVERAGE

| SUITE                                    | TOTAL | PASS | FAIL | STATUS      |
|------------------------------------------|-------|------|------|-------------|
| VAPT Remediation (V-001 - V-008)         | 8     | 8    | 0    | ✓ All Fixed |
| Unit Tests — lib & components            | 43    | 43   | 0    | ✓ PASS      |
| Regression Tests — full feature coverage | 25    | 25   | 0    | ✓ PASS      |

VAPT

## Security Vulnerability Assessment & Penetration Test — Remediation Status

8 findings · 8 fixed

V-001

### Hardcoded MongoDB Connection String

CRITICAL

✓ FIXED

Data Exposure · Database Layer ( `src/lib/db.ts` )

#### ORIGINAL FINDING

The MongoDB connection URI, including credentials and cluster hostname, was hardcoded in the source bundle — could extract live database credentials.

#### FIX APPLIED

Hardcoded URI removed from `db.ts`. Connection is now read exclusively from `process.env`.

V-002

### Missing Bearer Token Authentication on API Routes

CRITICAL

✓ FIXED

Broken Access Control · API Layer

#### ORIGINAL FINDING

All `/api/*` route handlers accepted requests without any authentication check. An

#### FIX APPLIED

New middleware module `src/lib/api-auth.ts` exports `requireAuth(request)` which

V-003

### ReDoS via Unescaped User Input in MongoDB Regex Queries

HIGH

✓ FIXED

Denial of Service · Search / Filter Endpoints

#### ORIGINAL FINDING

Search query parameters were interpolated directly into MongoDB `$regex` operator, causing backtracking, hanging the Node.js event loop.

#### FIX APPLIED

`escapeRegex(str: string)` added to `src/lib/validation.ts`. The function uses `escapeRegExp`

V-004

## Unbounded Pagination Parameters Causing Resource Exhaustion

HIGH

✓ FIXED

Denial of Service · Pagination Logic

### ORIGINAL FINDING

The `page` and `limit` query parameters were used without validation. A request with a large `limit` parameter would result in a single response, exhausting server memory and network bandwidth.

### FIX APPLIED

`safePagination(pageStr, limitStr)` added to `src/lib/validation.ts`. Page is cl

V-005

## Missing Input Validation Allowlists on Write Endpoints

HIGH

✓ FIXED

Injection / Mass Assignment · POST/PUT Handlers

### ORIGINAL FINDING

Request bodies were passed directly to Mongoose `.create()` / `.findByIdAndUpdate` without validation, allowing attackers to inject malicious data or bypass business rules.

### FIX APPLIED

Comprehensive Zod schema allowlists added in `src/lib/validation.ts`: `AssetCre`

V-006

## Missing HTTP Security Headers

MEDIUM

✓ FIXED

Defence-in-Depth · CDN / Static Hosting Layer

### ORIGINAL FINDING

Responses lacked `X-Frame-Options`, `X-Content-Type-Options`, `Strict-Transport-Security`, and `Content-Security-Policy`, leading to potential security leakage.

### FIX APPLIED

`public/_headers` file created for CDN delivery. Headers applied globally to all routes.

V-007

Weak Password Policy — No Minimum Length Enforced

MEDIUM

✓ FIXED

Authentication Weakness · Login & Register Forms

ORIGINAL FINDING

The login form accepted passwords of any length, including single-character inputs observer.

FIX APPLIED

Login form now enforces `password.length < 8` guard before any auth attempt, ret

V-008

Notification Read-State Not Persisted — Session Bleed

LOW

✓ FIXED

UX Security / Information Integrity · Notification Centre

ORIGINAL FINDING

Notification read/unread state was held only in React component state. On page refinance alert had been acknowledged, leading to repeated false-alarm urgency.

FIX APPLIED

Notification read-state is now persisted to `localStorage` under the key `gati_read`.

UNIT

Unit Test Results — Library Functions & UI Components

43 tests · 0 failures

2.1 ESCAPEREGEX() — SRC/LIB/VALIDATION.TS

| ID     | TEST NAME                 | INPUT / SCENARIO                            | EXPECTED RESULT                                  | STATUS |
|--------|---------------------------|---------------------------------------------|--------------------------------------------------|--------|
| UT-001 | Plain alphanumeric string | <code>escapeRegex("Toyota Fortuner")</code> | Returns <code>"Toyota Fortuner"</code> unchanged | ✓ PASS |
| UT-002 |                           | <code>escapeRegex("v1.0.2")</code>          |                                                  | ✓ PASS |

| ID     | TEST NAME                                   | INPUT / SCENARIO                                           | EXPECTED RESULT                                                                         | STATUS |
|--------|---------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|
|        | <b>String with dot metacharacter</b>        |                                                            | Returns<br><code>"v1\\.0\\.2"</code>                                                    |        |
| UT-003 | <b>String with asterisk</b>                 | <code>escapeRegex("a*b")</code>                            | Returns <code>"a\\*b"</code>                                                            | ✓ PASS |
| UT-004 | <b>ReDoS injection payload</b>              | <code>escapeRegex("(a+)+\$")</code>                        | Returns<br><code>"\\(a\\++\\)\\+\$"</code> — all metacharacters escaped                 | ✓ PASS |
| UT-005 | <b>Empty string</b>                         | <code>escapeRegex("")</code>                               | Returns <code>""</code> — no error thrown                                               | ✓ PASS |
| UT-006 | <b>All 12 metacharacters in one string</b>  | <code>escapeRegex(". * + ? ^ \$ { } ( )   [ ] \\"")</code> | Every character prefixed with a backslash; output safe for MongoDB <code>\$regex</code> | ✓ PASS |
| UT-007 | <b>Unicode &amp; whitespace passthrough</b> | <code>escapeRegex("भारत Motors")</code>                    | Returns string unchanged — no false escaping of non-meta chars                          | ✓ PASS |

## 2.2 SAFE PAGINATION() — SRC/LIB/VALIDATION.TS

| ID     | TEST NAME                               | INPUT / SCENARIO                          | EXPECTED RESULT                     | STATUS |
|--------|-----------------------------------------|-------------------------------------------|-------------------------------------|--------|
| UT-008 | <b>Valid page and limit</b>             | <code>safePagination("3", "25")</code>    | <code>{ page: 3, limit: 25 }</code> | ✓ PASS |
| UT-009 | <b>Null inputs use defaults</b>         | <code>safePagination(null, null)</code>   | <code>{ page: 1, limit: 20 }</code> | ✓ PASS |
| UT-010 | <b>Non-numeric strings use defaults</b> | <code>safePagination("abc", "xyz")</code> | <code>{ page: 1, limit: 20 }</code> | ✓ PASS |

| ID     | TEST NAME                          | INPUT / SCENARIO              | EXPECTED RESULT         | STATUS |
|--------|------------------------------------|-------------------------------|-------------------------|--------|
| UT-011 | Limit above ceiling clamped to 100 | safePagination("1", "999999") | { page: 1, limit: 100 } | ✓ PASS |
| UT-012 | Page below floor clamped to 1      | safePagination("0", "10")     | { page: 1, limit: 10 }  | ✓ PASS |
| UT-013 | Negative page clamped to 1         | safePagination("-5", "20")    | { page: 1, limit: 20 }  | ✓ PASS |
| UT-014 | Limit of exactly 1 accepted        | safePagination("2", "1")      | { page: 2, limit: 1 }   | ✓ PASS |

### 2.3 REQUIREAUTH() & ISAUTHERROR() — SRC/LIB/API-AUTH.TS

| ID     | TEST NAME                       | INPUT / SCENARIO                                                                 |
|--------|---------------------------------|----------------------------------------------------------------------------------|
| UT-015 | Missing Authorization header    | Request with no <code>Authorization</code> header                                |
| UT-016 | Header without "Bearer " prefix | <code>Authorization: Token abc123</code>                                         |
| UT-017 | Valid Bearer token — admin role | <code>Bearer &lt;base64({"email":"admin@gatitech.in","role":"admin"})&gt;</code> |
| UT-018 | Token with missing email field  | <code>Bearer &lt;base64({"role":"admin"})&gt;</code>                             |
| UT-019 | Token with missing role field   | <code>Bearer &lt;base64({"email":"x@x.com"})&gt;</code>                          |

| ID     | TEST NAME                                       | INPUT / SCENARIO                                              |
|--------|-------------------------------------------------|---------------------------------------------------------------|
| UT-020 | <b>Corrupted base64 token</b>                   | <code>Bearer !!!not-base64!!!</code>                          |
| UT-021 | <b>isAuthError — NextResponse is recognised</b> | <code>isAuthError(NextResponse.json({}, {status:401}))</code> |
| UT-022 | <b>isAuthError — plain object is rejected</b>   | <code>isAuthError({ userId: "x", role: "admin" })</code>      |

## 2.4 ZOD SCHEMAS — ASSETCREATESCHEMA, VEHICLECREATESCHEMA, MAINTENANCECREATESCHEMA

| ID     | TEST NAME                                         | INPUT / SCENARIO                                                           | EXPECTED RESULT                                                           | STATUS |
|--------|---------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|
| UT-023 | <b>AssetCreateSchema — valid minimal input</b>    | <code>{ name: "Generator", category: "equipment" }</code>                  | Parses successfully; optional fields absent from output                   | ✓ PASS |
| UT-024 | <b>AssetCreateSchema — missing required name</b>  | <code>{ category: "vehicle" }</code>                                       | ZodError with path <code>["name"]</code> , code <code>invalid_type</code> | ✓ PASS |
| UT-025 | <b>AssetCreateSchema — invalid category enum</b>  | <code>{ name: "X", category: "spaceship" }</code>                          | ZodError on <code>category</code> field — not in enum                     | ✓ PASS |
| UT-026 | <b>AssetCreateSchema — extra fields stripped</b>  | <code>{ name: "X", category: "tool", __proto__: {}, isAdmin: true }</code> | Extra keys silently stripped; parsed output contains only valid fields    | ✓ PASS |
| UT-027 | <b>AssetCreateSchema — name exceeds 200 chars</b> | name string of 201 characters                                              | ZodError on <code>name</code> — max length exceeded                       | ✓ PASS |

| ID     | TEST NAME                                              | INPUT / SCENARIO                                                                    | EXPECTED RESULT                                                            | STATUS |
|--------|--------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|
| UT-028 | <b>VehicleCreateSchema — valid full input</b>          | <pre>{ make: "Maruti",   model: "Swift",   year: 2024,   fuelType: "petrol" }</pre> | Parses successfully with all fields present                                | ✓ PASS |
| UT-029 | <b>VehicleCreateSchema — invalid fuelType enum</b>     | <pre>{ make: "X",   model: "Y",   year: 2020,   fuelType: "hydrogen" }</pre>        | ZodError on fuelType — not in enum [petrol, diesel, electric, hybrid, cng] | ✓ PASS |
| UT-030 | <b>VehicleCreateSchema — year below 1900</b>           | <pre>{ make: "X",   model: "Y",   year: 1850 }</pre>                                | ZodError on year — min 1900 not satisfied                                  | ✓ PASS |
| UT-031 | <b>MaintenanceCreateSchema — valid minimal</b>         | <pre>{ assetId: "abc123",   assetName: "Forklift" }</pre>                           | Parses successfully                                                        | ✓ PASS |
| UT-032 | <b>MaintenanceCreateSchema — missing assetId</b>       | <pre>{ assetName: "Forklift" }</pre>                                                | ZodError on assetId — required                                             | ✓ PASS |
| UT-033 | <b>MaintenanceCreateSchema — parts array too large</b> | parts array with 51 entries                                                         | ZodError on parts — max 50 entries                                         | ✓ PASS |

## 2.5 PLANS.TS — PLANS CONSTANT & PLANID TYPE

| ID     | TEST NAME                       | INPUT / SCENARIO                       | EXPECTED RESULT              |
|--------|---------------------------------|----------------------------------------|------------------------------|
| UT-034 | <b>PLANS has exactly 3 keys</b> | <code>Object.keys(PLANS).length</code> | 3 — starter, pro, enterprise |
| UT-035 | <b>Starter plan is free</b>     | <code>PLANS.starter.price</code>       |                              |



| ID     | TEST NAME                      | INPUT / SCENARIO                                                 | EXPECTED RESULT                                                                                     |
|--------|--------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|        |                                |                                                                  | 0 and<br>priceLabel<br>=== "Free"                                                                   |
| UT-036 | Pro plan limits                | PLANS.pro.limits                                                 | { assets:<br>500,<br>branches:<br>5, users:<br>10 }                                                 |
| UT-037 | Enterprise limits are Infinity | PLANS.enterprise.limits.assets                                   | Infinity                                                                                            |
| UT-038 | PlanId union covers all 3 keys | TypeScript: type PlanId =<br>"starter"   "pro"  <br>"enterprise" | Compile-<br>time type<br>check<br>passes;<br>assigning<br>any other<br>string<br>causes TS<br>error |

## 2.6 LOGIN FORM — SRC/APP/(AUTH)/LOGIN/PAGE.TSX

| ID     | TEST NAME                     | INPUT / SCENARIO                              | EXPECTED RESULT                                                                                        | STATUS |
|--------|-------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|
| UT-039 | Empty form submission blocked | Submit with email and password both empty     | HTML5<br>required<br>prevents<br>submit; no<br>API call<br>made                                        | ✓ PASS |
| UT-040 | Short password inline error   | Password = "abc"<br>(3 chars), valid<br>email | Error alert:<br>"Password<br>must be at<br>least 8<br>characters."<br>— no<br>network<br>request fired | ✓ PASS |
| UT-041 | Wrong credentials show error  | email:<br>wrong@test.com ,                    | Error:<br>"Invalid<br>email or                                                                         | ✓ PASS |

| ID     | TEST NAME                                      | INPUT / SCENARIO                                       | EXPECTED RESULT                                                                      | STATUS |
|--------|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|--------|
|        |                                                | password:<br>wrongpass1                                | password.<br>Please try again." — no redirect                                        |        |
| UT-042 | <b>Valid credentials redirect to dashboard</b> | email:<br>admin@gatitech.in ,<br>password:<br>admin123 | gati_auth<br>written to<br>localStorage;<br>router<br>pushes to /<br>dashboard       | ✓ PASS |
| UT-043 | <b>No demo credentials shown in UI</b>         | Inspect rendered login page DOM                        | No<br>hardcoded<br>demo<br>username or<br>password<br>visible in the<br>UI or source | ✓ PASS |

## 2.7 REGISTER FORM — SRC/APP/(AUTH)/REGISTER/PAGE.TSX

| ID     | TEST NAME                                          | INPUT / SCENARIO                  | EXPECTED RESULT                                                                          | STATUS |
|--------|----------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------|--------|
| UT-044 | <b>All fields empty — validation summary shown</b> | Submit form with all fields blank | Field-level error messages shown for orgName, fullName, email, password, confirmPassword | ✓ PASS |
| UT-045 | <b>Invalid email format rejected</b>               | email:<br>"notanemail"            | Error: "Please enter a valid email address." under email field                           | ✓ PASS |
| UT-046 | <b>Password shorter than 8 chars rejected</b>      | password:<br>"pass"               | Error: "Password must be at least 8 characters." under password field                    | ✓ PASS |
| UT-047 | <b>Password mismatch rejected</b>                  | password:<br>"securepass" ,       | Error: "Passwords do not match." under                                                   | ✓ PASS |

| ID     | TEST NAME                          | INPUT / SCENARIO                                          | EXPECTED RESULT                                                                 | STATUS |
|--------|------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------|--------|
|        |                                    | confirm:<br>"different1"                                  | confirmPassword field                                                           |        |
| UT-048 | Plan selection persisted on submit | Select Pro plan, fill all valid fields, submit            | localStorage<br>gati_auth.plan<br>=== "pro" ;<br>router redirects to /dashboard | ✓ PASS |
| UT-049 | Default plan is Starter            | Load register page without interacting with plan selector | Starter card has selected ring applied;<br>selectedPlan<br>=== "starter"        | ✓ PASS |

## 2.8 PRICING PAGE — SRC/APP/PRICING/PAGE.TSX

| ID     | TEST NAME                          | INPUT / SCENARIO                                        | EXPECTED RESULT                                                          | STATUS |
|--------|------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------|--------|
| UT-050 | All 3 plan cards render            | Render / pricing                                        | Starter, Pro, and Enterprise cards all present in DOM                    | ✓ PASS |
| UT-051 | Monthly pricing shown by default   | Initial render — toggle state<br>annual<br>===<br>false | Pro shows "₹2,999/mo"; Enterprise shows "₹9,999/mo"; no "Save 20%" badge | ✓ PASS |
| UT-052 | Annual toggle applies 20% discount | Click annual toggle                                     | Pro shows "₹2,399/mo"; Enterprise shows "₹7,999/mo"; "Save               | ✓ PASS |

| ID     | TEST NAME                                  | INPUT / SCENARIO      | EXPECTED RESULT                                                                                    | STATUS |
|--------|--------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------|--------|
|        |                                            |                       | 20%" badge appears                                                                                 |        |
| UT-053 | <b>Starter remains Free in annual mode</b> | Toggle annual billing | Starter plan still shows "Free" — price === 0 never discounted                                     | ✓ PASS |
| UT-054 | <b>Pro card marked "Most Popular"</b>      | Inspect plan cards    | Pro card has highlight: true ; "Most Popular" badge renders; card is visually elevated (scale-105) | ✓ PASS |

## 2.9 SIDEBAR PLAN BADGE & SETTINGS SUBSCRIPTION TAB

| ID     | TEST NAME                                     | INPUT / SCENARIO                                                     | EXPECTED RESULT                                              | STATUS |
|--------|-----------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------|--------|
| UT-055 | <b>Sidebar shows Starter badge by default</b> | <code>gati_auth</code> absent from localStorage                      | Sidebar badge shows "Starter Plan" with emerald colour class | ✓ PASS |
| UT-056 | <b>Sidebar badge updates to Pro</b>           | <code>gati_auth.plan = "pro"</code> set in localStorage before mount | Sidebar badge shows "Pro Plan" with orange colour class      | ✓ PASS |
| UT-057 | <b>Sidebar badge updates to Enterprise</b>    | <code>gati_auth.plan = "enterprise"</code> set in localStorage       | Sidebar badge shows "Enterprise"                             | ✓ PASS |

| ID     | TEST NAME                             | INPUT / SCENARIO                         | EXPECTED RESULT                                                                           | STATUS |
|--------|---------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------|--------|
| UT-058 | Subscription tab renders usage meters | Navigate to Settings → Subscription tab  | with violet colour class                                                                  | ✓ PASS |
|        |                                       |                                          | Three usage progress bars shown: Assets, Users, Branches — each with current/limit values |        |
|        |                                       |                                          |                                                                                           |        |
| UT-059 | Billing history table renders         | Subscription tab billing history section | Mock billing rows shown with date, description, amount, and invoice download link         | ✓ PASS |
| UT-060 | Upgrade Plan link points to / pricing | Subscription tab "Upgrade Plan" button   | <code>href="/pricing"</code> — navigates to pricing page                                  | ✓ PASS |

REGRESSION

Regression Test Results — Previously Working Feature Coverage

25 tests · 0 regressions

| ID     | FEATURE AREA | TEST DESCRIPTION                                                                           | EXPECTED BEHAVIOUR                                                     | STATUS |
|--------|--------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------|
| RT-001 | Auth Guard   | Access <code>/dashboard</code> without <code>gati_auth</code> in <code>localStorage</code> | Middleware redirects to <code>/login</code> ; dashboard never rendered | ✓ PASS |

| ID     | FEATURE AREA   | TEST DESCRIPTION                                                       | EXPECTED BEHAVIOUR                                                                                    | STATUS |
|--------|----------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|
| RT-002 | Auth Guard     | Authenticated user visits <code>/login</code>                          | Redirect to <code>/dashboard</code> — already logged in                                               | ✓ PASS |
| RT-003 | Dashboard Load | Authenticated user loads <code>/dashboard</code>                       | KPI cards, vehicle counts, maintenance alerts, and finance summary all render without errors          | ✓ PASS |
| RT-004 | Vehicles CRUD  | Navigate to <code>/dashboard/vehicles</code> , open Add Vehicle dialog | Vehicle form renders all fields; submit creates entry in list; row count increments by 1              | ✓ PASS |
| RT-005 | Vehicles CRUD  | Edit existing vehicle — change model name, save                        | Updated model name reflected in table row; no duplicate entry created                                 | ✓ PASS |
| RT-006 | Vehicles CRUD  | Delete vehicle — confirm dialog, proceed                               | Vehicle removed from list; row count decrements; no console errors                                    | ✓ PASS |
| RT-007 | Assets CRUD    | Add new asset with category equipment                                  | Asset appears in list with correct category chip; status defaults to active                           | ✓ PASS |
| RT-008 | Assets CRUD    | Edit asset — update purchase price and condition                       | Updated values saved and reflected immediately; depreciation display recalculates                     | ✓ PASS |
| RT-009 | Maintenance    | Create new maintenance record with priority critical                   | Record appears in maintenance list with red critical badge; alert counter in sidebar badge increments | ✓ PASS |

| ID     | Feature Area      | Test Description                                         | Expected Behaviour                                                                              | Status |
|--------|-------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| RT-010 | Maintenance       | Mark maintenance record as completed                     | Status changes to completed; record moves out of pending queue; cost fields lock                | ✓ PASS |
| RT-011 | Parts & Inventory | Navigate to <code>/dashboard/parts</code> , add new part | Part entry created with SKU, quantity, and unit price; stock total updates in header            | ✓ PASS |
| RT-012 | Parts & Inventory | Edit part quantity — simulate stock receipt              | New quantity saved; low-stock warning clears if quantity now above threshold                    | ✓ PASS |
| RT-013 | Customers         | Add new customer with email and phone                    | Customer record created; appears in customer list; linked purchases count shows 0               | ✓ PASS |
| RT-014 | Customers         | Delete customer — confirmation required                  | Customer removed from list; associated records show unlinked state; no orphan data errors       | ✓ PASS |
| RT-015 | Vendors           | Add new vendor with GST number and category              | Vendor record saved; appears in vendor list with correct category tag                           | ✓ PASS |
| RT-016 | Finance           | Load <code>/dashboard/finance</code>                     | Revenue, expenses, and profit charts render; KPI cards display ₹ values with correct formatting | ✓ PASS |
| RT-017 | Finance           | Filter finance by month — June 2026                      | Charts and table update to show only June 2026 transactions; totals recalculate                 | ✓ PASS |
| RT-018 | Reports           | Navigate to <code>/dashboard/reports</code> ,            | Report renders with asset count by category, total value, depreciation summary                  | ✓ PASS |

| ID     | FEATURE AREA                 | TEST DESCRIPTION                          | EXPECTED BEHAVIOUR                                                                                                                               | STATUS |
|--------|------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|        |                              | generate asset report                     |                                                                                                                                                  |        |
| RT-019 | Settings — Profile Tab       | Update display name in Settings → Profile | Name saved to state; success toast shown; sidebar greeting updates on next render                                                                | ✓ PASS |
| RT-020 | Settings — Notifications Tab | Toggle email notification preference off  | Toggle state persists after tab switch; no revert on navigation                                                                                  | ✓ PASS |
| RT-021 | Logout                       | Click Logout in sidebar                   | <code>gati_auth</code> removed from <code>localStorage</code> ; router redirects to <code>/login</code> ; back button does not reenter dashboard | ✓ PASS |
| RT-022 | Notification Read-State      | Mark 2 alerts as read, refresh the page   | Both alerts remain read after refresh; unread count in top-nav badge decrements correctly; persisted in <code>gati_read_notifications</code> key | ✓ PASS |
| RT-023 | Notification Read-State      | Click "Mark all read"                     | All alerts marked read; badge count reaches 0; state persisted to <code>localStorage</code>                                                      | ✓ PASS |
| RT-024 | Responsive Layout            | Resize viewport to 375px (mobile)         | Sidebar collapses; hamburger menu visible; all nav items accessible via mobile drawer; no horizontal overflow                                    | ✓ PASS |
| RT-025 | Responsive Layout            | Resize viewport to 768px (tablet)         | Sidebar renders in icon-only compact mode; tooltips appear on hover; main content area fills remaining width correctly                           | ✓ PASS |



|                                |                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------|
| <b>Framework</b>               | Next.js 15 (App Router, static export + API routes)                                    |
| <b>Runtime</b>                 | Node.js 20 LTS / Edge Runtime (middleware)                                             |
| <b>Unit Test Runner</b>        | Vitest 1.x + React Testing Library 14                                                  |
| <b>E2E / Regression Runner</b> | Playwright 1.44 — Chromium, Firefox, WebKit                                            |
| <b>Validation Library</b>      | Zod 3.x                                                                                |
| <b>Database</b>                | MongoDB Atlas (staging cluster) via Mongoose 8                                         |
| <b>VAPT Tooling</b>            | OWASP ZAP 2.14, Burp Suite Professional 2024.3, manual code review                     |
| <b>CI/CD</b>                   | GitHub Actions — lint → typecheck → unit tests → build → Playwright on each PR to main |

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